

DP-2026-001-09-vehicle-parameterized-gnc

One guidance-navigation-control step function parameterized by vehicle physics across media

Defensive publication. This document is published solely to place its subject matter into the public domain of technical knowledge as prior art. **No patent protection is sought by the author for any subject matter described here.**

Author / declarant: Jun Kawasaki (root@junkawasaki.com) **Disclosure set:** DP-2026-001, part 9 of 10 · first published 2026-08-19 (UTC) **Enabling reference implementation:** public source code under <https://github.com/kotoba-lang/>, licensed Apache-2.0, fixed by commit and tree SHA-1 in the evidence bundle at <https://github.com/com-junkawasaki/prior-art> — that bundle carries RFC 3161 time-stamp tokens and OpenTimestamps attestations over the fixing manifest.

Note on the "Variants and generalizations" section below

The variants are stated deliberately and at length. They are disclosed so that a later filing directed to an obvious variation of the mechanism is met by an express **written** disclosure of that variation, and not merely by an argument that the variation would have been obvious. Where the text says "any", the word is intended in its ordinary broad sense and the enumeration following it is illustrative, not exhaustive.

Working source code implementing the mechanism is publicly available at the commits fixed in the evidence bundle. A person of ordinary skill in robotics software can read, build and run that code; this document is therefore an enabling disclosure and not a mere statement of a result.

Problem

Ground, marine, and aerial autonomy are built as separate stacks. The guidance and decision logic is duplicated three times and diverges, although the difference between the vehicles is confined to the physics of how a command becomes motion.

Mechanism

A **single** guidance, navigation and control step function — goal handling, path planning, path tracking, clearance checking, stuck detection, arrival detection, and telemetry — parameterized by a **vehicle model value** supplying only the medium-specific relations:

- a ground model (bicycle kinematics with steering and wheelbase);
- a marine model (hull hydrodynamic resistance and thrust);
- a fixed-wing model (stall speed, bank-limited turn radius, air-density lapse with altitude);
- a multirotor model (thrust-to-weight, tilt-limited acceleration).

The step function is the same code in every case; the vehicle model is data. A fleet may therefore be heterogeneous and still be commanded, verified, and audited by one implementation, and a new vehicle class is added by supplying a model rather than by writing a stack.

The same function supports fleet-level operation: multiple agents each with pose and goal over a shared static scene, with an "ahead and closing" predicate used for mutual yielding, a minimum-separation query over the fleet, and an all-arrived predicate.

Variants and generalizations

- Vehicle classes may further include: differential-drive; omnidirectional and mecanum; tracked; Ackermann with trailer; articulated and multi-body machines; legged; underwater vehicles with buoyancy and added mass; airships; spacecraft with reaction control; hybrid amphibious vehicles.
- The parameterization may be: a record of coefficients; a function value; a table; an identified model fitted from logged data; a learned dynamics model; a differentiable model.
- The shared logic may extend to: mission sequencing; energy-aware routing; contingency and return-to-home; formation keeping; traffic rules and right-of-way; collision-avoidance protocol; degraded-mode behaviour.
- Heterogeneous fleets may be coordinated by: pairwise yielding predicates; priority ordering; auction or market assignment; centralized scheduling; the precomputed verified choreography of Section 3.
- The arrangement applies to simulation and to real vehicles with the same code, and, with Section 6, to a governor predicting a proposed motion's consequence before admitting it.

Reference implementation

`kotoba-lang/kami-autodrive` (`bicycle-model`, `ship-hydro`, `stall-speed`, `fixed-wing`, `multirotor`, `isa-density`, `limits`, `step`, `step-multimodal`, `fleet-agent`, `new-fleet`, `ahead-and-closing?`, `min-separation`, `all-arrived?`, `autopilot-config`, `new-autopilot`, `set-goal`, `telemetry`).

中文摘要 / Chinese abstract

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一个制导-导航-控制步进函数由载具物理参数化，跨越介质：目标处理、规划、跟踪、净空检查、卡滞检测、到达判定与遥测为同一份代码，仅以载具模型（地面自行车模型、船舶水动力、固定翼失速与坡度限转弯、多旋翼推重比与倾角限加速度）作为数据。因此异构机群可由一份实现指挥、验证与审计，新增载具类别只需提供模型。

上文“变型与推广”一节系**有意穷举列举**，以使针对本机制之显而易见变化的在后申请面对的是明确的书面公开，而非仅仅是“显而易见”的主张。

Statement of dedication

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